

Highlights

Federated Digital Twins for Cyber-Physical Systems-of-Systems: A Taxonomy of Data, Model, and Control Federation

Sashko Ristov, Marco Cotrotzo

- Federated digital twins are defined and positioned within systems-of-systems.
- A seven-dimensional taxonomy for federated digital twins is introduced.
- Architectural and interaction semantics of federation are systematically distinguished.
- A microgrid case study demonstrates the applicability of the taxonomy in a new prototype library.
- The taxonomy provides a foundation for future autonomous digital twin ecosystems.

Federated Digital Twins for Cyber-Physical Systems-of-Systems: A Taxonomy of Data, Model, and Control Federation

Sashko Ristov^{a,*}, Marco Cotrotzo^a

^aUniversity of Innsbruck, Innrain 52, Innsbruck, 6020, Austria

Abstract

Digital twins are increasingly used to monitor, analyze, predict, and optimize cyber-physical and industrial systems. At the same time, modern cyber-physical infrastructures are evolving toward interconnected *systems-of-systems* in which multiple digital twins must coordinate information, predictions, and control decisions across organizational and technical boundaries. Although the notion of interconnection is increasingly emerging in both research and industrial practice, the term *federation* remains ambiguous and is used to describe substantially different forms of data sharing, hierarchical composition, interoperability, runtime coordination, and distributed optimization.

This paper presents a taxonomy of *federated digital twins* for cyber-physical and industrial systems. Unlike existing classifications that primarily distinguish structural forms of multi-twin systems, the proposed taxonomy focuses on federation semantics and explicitly characterizes how twins cooperate, exchange information, and coordinate decisions. The taxonomy distinguishes seven dimensions organized into architectural and interaction dimensions: integration level, federation direction, twin abstraction level, runtime composition, federation depth, federation awareness, and exchange mode. A key contribution of the taxonomy is the explicit distinction between digital-twin abstraction layers and the federation depth through which these layers participate in federation. To demonstrate the applicability of the taxonomy, the paper presents a prototype realization of a vertically federated microgrid digital twin integrating photovoltaic, battery, and electric vehicle charging system-level twins. The prototype operationalizes all seven dimensions of the proposed taxonomy through executable hierarchical federation of PV, battery, and EV twins.

Overall, the proposed taxonomy provides a conceptual foundation for future research and engineering of federated digital twins in large-scale cyber-physical and industrial ecosystems, in particular dynamic runtime federation, federation-aware coordination, negotiation-based exchange, predictive-model federation, and large-scale decentralized cyber-physical ecosystems.

Keywords: federation, digital twins, microgrid, taxonomy, system of systems, models, cyber physical systems

1. Introduction

Digital twins have become a central paradigm for monitoring, analyzing, predicting, and optimizing the behavior of cyber-physical and industrial systems [1, 2]. By combining real-time operational data with executable models and control logic, digital twins enable advanced operational capabilities across domains such as manufacturing, energy systems, transportation, smart buildings, and industrial automation [3, 4, 5, 6]. Recent advances in Industrial Internet of Things (IIoT), edge and cloud computing, artificial intelligence, and distributed cyber-physical infrastructures [5, 6] have further accelerated the adoption of digital twins as a foundation for intelligent and autonomous system operation.

At the same time, cyber-physical systems are increasingly evolving from isolated systems toward interconnected *systems-of-systems* composed of heterogeneous, autonomous, and independently operated subsystems [4, 7]. Such systems-of-systems environments emerge in domains including manufac-

turing, smart cities, energy systems, transportation, and health-care, where multiple systems must collaborate while preserving their operational independence. As a consequence, digital twins are no longer confined to representing individual assets or standalone systems. Instead, they must support coordination, information exchange, and decision-making across larger ecosystems spanning multiple technical, organizational, and geographical domains. The concepts of systems and systems-of-systems, and their digital representations are introduced in **Section 2**.

While digital twins provide powerful capabilities for monitoring, analysis, prediction, and optimization, individual twins are typically designed around a single system boundary. In systems-of-systems environments, however, higher-level objectives often emerge from the interactions among multiple systems. Consequently, isolated digital twins are insufficient to support ecosystem-level coordination, optimization, and decision-making. This challenge is illustrated through a microgrid running example in **Section 3**, where photovoltaic (PV) generation, battery storage, and electric vehicle (EV) charging systems must cooperate to achieve additional system-level objectives.

*Corresponding author

Email addresses: sashko.ristov@uibk.ac.at (Sashko Ristov), marco.cotrotzo@uibk.ac.at (Marco Cotrotzo)

To support collaboration across such systems-of-systems, recent research has introduced the concept of *Federated Digital Twins*. Federated digital twins enable multiple digital twins to exchange information, coordinate behavior, and jointly represent larger cyber-physical environments that span organizational, operational, and domain boundaries [8, 9]. Building on this vision, the Internet of Federated Digital Twins has been proposed as a large-scale framework for integrating fragmented and heterogeneous digital twins into unified ecosystems capable of delivering societal-scale services [10]. Recent research has further started addressing operational and computational challenges of federation itself, including heterogeneity in processing capabilities and coordination efficiency among participating twins [11]. Despite these advances, the notion of *federation* remains ambiguous and is frequently used to describe substantially different concepts. Existing approaches use federation to denote semantic interoperability [12], data sharing and exchange [13], distributed deployment and execution [14], hierarchical composition of digital twins [15], collaborative optimization and ecosystem-level coordination [16, 17], and federated learning or distributed intelligence. Consequently, studies often use the same terminology to describe fundamentally different architectural and interaction patterns. This ambiguity makes it difficult to compare existing approaches, position new contributions, and reason systematically about the design of federated digital twin architectures. **Section 4** analyzes representative approaches and highlights these different interpretations of federation in greater detail.

To address this challenge and research gap, this paper presents a taxonomy of federated digital twins for cyber-physical and industrial system-of-systems. The taxonomy is developed from the interdisciplinary foundations of systems-of-systems, digital twin abstraction layers, and federation concepts from distributed systems engineering. It distinguishes federation along seven orthogonal dimensions organized into two groups. The architectural dimensions capture the structural organization of federated twins, including the integration level of the represented systems, the federation direction, the participating twin abstraction levels, and the runtime composition of the federation. The interaction dimensions characterize how participating twins cooperate, including the federation depth, federation awareness, and exchange mode. Together, these dimensions provide a systematic framework for distinguishing different forms of federation and for reasoning about the design of federated digital twin architectures and interactions within and between them. The taxonomy construction process is presented in **Section 5**, while the resulting taxonomy is introduced in **Section 6**.

To demonstrate the applicability of the proposed taxonomy, the paper presents a prototype realization of a federated microgrid digital twin integrating PV generation, battery storage, and EV charging systems. The prototype is used to illustrate how the seven taxonomy dimensions can be applied to characterize a concrete federation architecture. In particular, the implementation demonstrates a vertically federated architecture in which system-level twins contribute to a higher-level microgrid twin through digital-shadow and prescriptive-model federation. The

prototype and its classification according to the proposed taxonomy are presented in **Section 7**.

The main contributions of this paper are:

- We conceptualize federated digital twins as a mechanism for enabling cooperation among digital twins in cyber-physical systems-of-systems and distinguish federation from related concepts such as aggregation, composition, interoperability, and ecosystem coordination.
- We introduce a seven-dimensional taxonomy of federated digital twins that systematically characterizes federation semantics through architectural and interaction dimensions, including integration level, federation direction, twin abstraction level, runtime composition, federation depth, federation awareness, and exchange mode.
- We demonstrate the applicability of the taxonomy through a prototype library and a federated microgrid digital twin integrating PV generation, battery storage, and EV charging systems.
- We identify emerging research directions for federated digital twins, including dynamic runtime federation, federation-aware coordination, negotiation-based interaction, predictive-model federation, and large-scale autonomous digital twin ecosystems.

While the proposed taxonomy provides a structured framework for describing federated digital twin architectures, it primarily focuses on technical federation semantics and does not explicitly capture governance structures, trust relationships, ownership boundaries, or contractual aspects of federation. Furthermore, dynamic federation mechanisms involving runtime discovery, negotiation, and autonomous formation of federations remain open research challenges. These limitations, together with directions for future research on federated digital twins, are discussed in **Section 8**. Despite these limitations, the proposed taxonomy establishes a conceptual foundation for systematically describing, comparing, and engineering federated digital twin architectures across cyber-physical systems and systems-of-systems and can be further extended toward federation-aware, autonomous, and large-scale digital twin ecosystems, as concluded in **Section 9**.

2. Foundations

This section introduces the conceptual foundations used to construct the taxonomy of federated digital twins. We first distinguish modules, systems, and systems-of-systems as integration levels commonly used in cyber-physical and industrial systems. We then summarize the role of digital twins as digital representations of these systems, with emphasis on their data, descriptive, predictive, and prescriptive abstraction levels. These concepts provide the basis for the running example and the taxonomy introduced later in the paper.

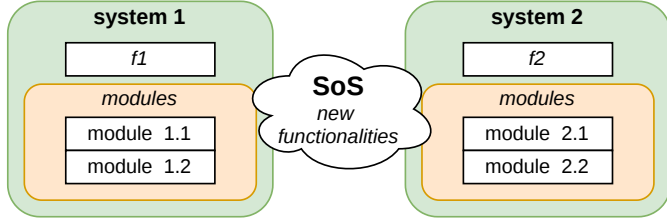


Figure 1: Conceptualizing system-of-systems.

2.1. From Modules to Systems and Systems-of-Systems

Cyber-physical systems are commonly structured across multiple integration levels, ranging from individual technical components to coordinated collections of independently operated systems. Understanding these levels is important because digital twins may represent entities at different scales, from sensing and actuation modules to complete systems and systems-of-systems.

In general, cyber-physical reference architectures and frameworks distinguish several layers below the module level, including devices, embedded components, chips, and communication elements [7]. However, the focus of this paper is not on the internal physical realization of hardware assets, but on operational functionalities and their digital representations. Therefore, we abstract lower-level physical and electronic details into modules that expose sensing, actuation, communication, or local control functionality.

Figure 1 illustrates the general transition from modules and systems toward systems-of-systems. At the lowest level considered in this paper are the *modules*, which provide specific technical functions such as sensing, actuation, or local control. A power meter (sensor), inverter (actuator), plug-detection sensor, or battery management controller can be treated as a module. Modules may expose data interfaces and operational states, but they usually do not provide the full functionality expected from a cyber-physical system.

A *system* is a composition of modules that jointly provide a well-defined operational functionality. Systems integrate sensing, actuation, communication, and local control into a coherent cyber-physical behavior visible at the operational level. For example, an EV charger combines plug-detection sensors, power metering, charging actuators, and local control logic to charge a connected vehicle with variable power. A battery system combines state-of-charge estimation, power conversion, metering, and safety control to store and release electrical energy. Similarly, a PV system combines generation, inversion, metering, and control to inject or curtail electrical power. Compared to individual modules, systems expose higher-level functionalities and operational objectives. They therefore form a natural abstraction boundary for monitoring, optimization, decision support, and automation. At this level, the internal implementation details of individual modules are partially encapsulated, while the externally visible behavior of the system becomes the primary concern.

A *system-of-systems* (SoS) is a composition of systems that retain a degree of operational and managerial independence

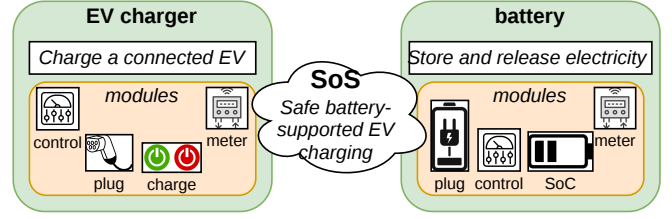


Figure 2: Battery-EV charger system-of-systems to extend the functionalities of the individual systems *EV charger* (charge a connected EV) *battery* (store and release electricity) into "safe battery-supported EV charging".

while jointly contributing to higher-level functionality. In contrast to monolithic systems, the participating systems in an SoS may continue operating independently, maintain local control strategies, and evolve separately over time. The coordination among systems introduces emergent functionalities that cannot be achieved by the individual systems alone.

In the energy domain, examples of SoS include microgrids, energy communities, distributed flexibility aggregators, and coordinated multi-site energy infrastructures. For instance, a microgrid may combine PV systems, stationary batteries, EV chargers, and grid interfaces into a coordinated energy ecosystem. While each constituent system continues to provide its own local functionality, the overall SoS may introduce higher-level objectives such as maximizing self-consumption, reducing peak demand, coordinating flexibility, or optimizing energy exchange with the grid.

Figure 2 presents an illustrative example in which a battery system and an EV charging system jointly form a simple system-of-systems. Individually, the EV charger provides the functionality of charging a connected vehicle, while the battery system provides the functionality of storing and releasing electricity. When coordinated together, the resulting SoS introduces the new emergent functionality of safe battery-supported EV charging.

2.2. Digital Twins in Cyber-Physical and Industrial Systems

Digital twins have become a central concept for engineering, monitoring, and optimizing cyber-physical and industrial systems. A digital twin is commonly understood as a digital representation of a physical or cyber-physical entity that is connected to the entity through data flows and used to support analysis, prediction, decision making, or control [1]. In industrial and energy domains, digital twins are used for asset monitoring, predictive maintenance, simulation, optimization, operator support, and system-level decision making.

A useful distinction in the digital twin literature is the separation between digital models, digital shadows, and digital twins [1]. A digital model represents a physical entity without automatic data synchronization. A digital shadow receives data from the physical entity, but the interaction is primarily one-directional from the physical to the digital representation. A digital twin extends this relation by enabling bidirectional interaction, where the digital representation can also influence the physical system through recommendations, setpoints, or control actions. This distinction is important for federation because

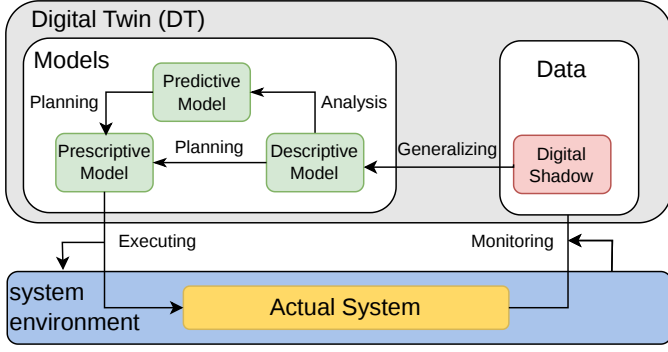


Figure 3: Conceptual framework for DTs based on [2], used later to motivate twin abstraction levels and federation depth.

different architectures may federate only data shadows, while others federate predictive models, prescriptive models, or control decisions.

Digital twins can also be understood through different abstraction levels, as described in Fig. 3. At the data level (*digital shadow*), a twin maintains a synchronized representation of measurements, events, configurations, and operational states. For example, an EV charger twin may maintain charging-session information, plug status, instantaneous charging power, and energy consumption measurements, while a battery twin may maintain state-of-charge values, charging/discharging power, and historical energy-flow data. At the descriptive level, the twin interprets these data to derive meaningful current-state information (*descriptive model*). For instance, a battery twin may determine whether the battery is currently available for charging or discharging, while an EV charger twin may estimate charging-session progress. At the predictive level, the twin estimates future behavior using simulation, historical observations, forecasts, or learned models (*predictive model*). Examples include forecasting PV generation, estimating future battery state of charge, or predicting EV charging demand and completion time. Finally, at the prescriptive level, the twin recommends or decides actions such as charging schedules, charging/discharging setpoints, battery utilization strategies, or PV curtailment decisions (*prescriptive model*).

3. Motivation with a Running Example: A Microgrid System-of-Systems

To motivate the need for federated digital twins, this section introduces a running example based on a microgrid system-of-systems. We first describe the individual systems and their corresponding digital twins. We then discuss why isolated digital twins are insufficient for system-level optimization and motivate the need for federation.

3.1. System-Level Twins in a Microgrid

The running example considered in this paper is a microgrid composed of three interacting cyber-physical systems: PV, stationary battery storage, and EV charger. The example is intentionally simple but representative of modern cyber-physical

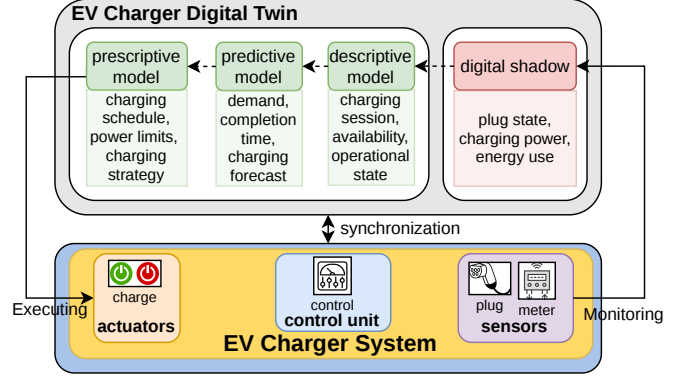


Figure 4: EV charger system and its modules (actuators, sensors, control unit), together with the respective EV charger digital twin, including respective digital shadow and descriptive, predictive, and prescriptive models.

energy systems, where independently operated systems must coordinate to achieve higher-level objectives such as maximizing self-consumption, coordinating flexibility, and optimizing energy flows. Each system provides its own local operational functionality and is represented by a corresponding system-level digital twin. The systems are intentionally modeled at the operational level rather than at the level of detailed physical realization. The focus is therefore placed on sensing, actuation, operational states, predictions, and control decisions relevant for energy management and coordination.

3.1.1. EV Charger System and Digital Twin

The EV charging system is responsible for charging connected EVs with variable charging power while respecting operational and safety constraints. At the operational level, the system combines sensing, metering, actuation, and local control functionality. Figure 4 illustrates the EV charging system together with its corresponding digital twin. The physical (actual) system consists of sensing modules for detecting plug status and charging conditions and metering modules for monitoring charging power and energy consumption; the actuator module comprises charging actuator; while a local control unit is responsible for charging execution.

The EV charger digital twin maintains synchronized operational data through its digital shadow and additionally provides descriptive, predictive, and prescriptive models. The descriptive model derives operational information such as charging-session state and charger availability. The predictive model estimates future charging demand, charging completion time, and expected charging behavior. Finally, the prescriptive model recommends or decides charging schedules, charging power limits, and charging strategies, and executes them to the actuators.

3.1.2. Battery System and Digital Twin

The battery system is responsible for storing and releasing electrical energy while respecting operational, safety, and degradation constraints. At the operational level, the system combines sensing, metering, charging/discharging actuation, and local control functionality.

The corresponding battery digital twin maintains operational states such as state of charge, charging/discharging power, energy-flow history, and operational availability through its digital shadow. The descriptive model determines whether the battery is currently available for charging or discharging and derives operational conditions from synchronized measurements. The predictive model estimates future state-of-charge evolution, charging/discharging behavior, and battery utilization trends. Finally, the prescriptive model may recommend charging/discharging schedules, battery utilization strategies, and operational power limits.

3.1.3. PV System and Digital Twin

The PV system is responsible for converting solar energy into electrical power and injecting this power into the local microgrid or external grid infrastructure. The system combines sensing, metering, inversion, and local control functionality.

The corresponding PV twin maintains synchronized operational data such as generated power, inverter state, energy production, and environmental measurements through its digital shadow. The descriptive model derives the current operational condition of the PV system and determines generation availability. The predictive model estimates future PV generation based on historical observations and weather forecasts. Finally, the prescriptive model may recommend generation strategies, inverter operating conditions, or PV curtailment decisions.

3.2. Why Isolated Twins Are Insufficient

Although the individual digital twins introduced in the previous subsection can independently monitor, predict, and optimize the behavior of their corresponding systems, they do not operate in isolation. The PV system, battery system, and EV charging system remain electrically and operationally coupled through shared energy flows, grid interaction, and local operational constraints. For example, the charging behavior of the EV charger directly affects battery utilization and local energy demand. Similarly, PV generation influences battery charging opportunities and the amount of energy imported from or exported to the grid. Consequently, local optimization performed independently by each twin may lead to conflicting or suboptimal behavior at the higher microgrid system-level. An EV charger may initiate charging when local PV generation is low, a battery may discharge without considering future charging demand, or a PV system may inject energy into the grid although local storage or charging flexibility is available.

These interactions motivate the need for *federated digital twins* capable of coordinating data, models, predictions, and control decisions across multiple systems. However, the notion of federation itself remains ambiguous. In some cases, federation refers to the exchange of monitored data or digital shadows among twins. In other cases, it refers to sharing descriptive or predictive models, distributed optimization, hierarchical control structures, event-driven coordination, or the creation of higher-level twins representing coordinated systems-of-systems.

The ambiguity becomes particularly visible in the microgrid example. A battery twin and an EV charger twin may exchange only operational measurements while preserving fully

local control strategies. Alternatively, the twins may coordinate charging and discharging decisions through distributed optimization. In another scenario, the PV, battery, and EV charger twins may become part of a parent microgrid twin that introduces new system-level objectives such as maximizing self-consumption or coordinating flexibility. These cases are fundamentally different from architectural, operational, and control perspectives, yet they are often all described simply as federated digital twins.

A second challenge concerns the depth and mode of interaction among twins. Some federations exchange only raw measurements or synchronized states, while others exchange predictions, constraints, flexibility descriptions, or control decisions. Similarly, interaction may rely on shared data spaces, pull-based access, event-driven communication, or negotiation-based coordination. Existing discussions on federation typically do not distinguish these aspects systematically.

These observations motivate the need for a systematic classification framework capable of distinguishing different forms of federated digital twins. To better understand how federation has been conceptualized in the literature and to identify the remaining research gaps, the next section reviews existing work on systems of digital twins and federated digital twins.

4. Related Work

This section reviews the recent work on systems of digital twins and federated digital twins. We focus on multiple interacting twins and the different views of federation.

4.1. Systems of Digital Twins

Recent work has started to systematize architectures composed of multiple digital twins. Gervasi et al. [18] present a systematic-narrative review of systems of digital twins and show that the literature is characterized by non-standardized and partly inconsistent terminology. Their review includes 62 studies and distinguishes five main typologies: Digital Twin Aggregate, Digital Twin Network, Federated Digital Twin, Composite Digital Twin, and Digital Twin Ecosystem. They further propose a formal framework for a Global Digital Twin, including a purpose-oriented representation, a projection operator, and a model-based equivalence relation for integrating heterogeneous twins. This work is highly complementary to ours. Gervasi et al. [18] primarily classify structural forms of systems of digital twins: whether multiple twins are aggregated, networked, composed, federated, or embedded in a broader ecosystem. In contrast, our taxonomy focuses on the federation semantics that may occur within or across such structures. We distinguish not only that multiple twins are connected, but also how they are connected, which abstraction layers participate in federation, whether the composition is static or dynamic, whether twins are aware of the federation, and how information, models, or control decisions are exchanged.

Several other works study related structural forms of multi-twin architectures. Wu et al. [19] define digital twin networks as many-to-many mappings among multiple one-to-one twins,

while other network-oriented works model networks of twins that communicate with one another or introduce higher-order twins responsible for coordinating edge or worker twins. Composite digital twins are studied by Larsen et al. [20], who describe how existing twins can be composed to construct higher-level twins or reusable DT services. Digital twin ecosystem approaches, such as those by Ricci [16], Calvo-Bascones [21], and Martinelli [17], emphasize open-ended collections of interconnected twins, services, and platforms that may be developed by different organizations and exploited in synergy within shared application scenarios.

4.2. Federated Digital Twins

Federated digital twins are increasingly emerging as a promising paradigm for connecting heterogeneous digital twins across organizational, operational, and domain boundaries. Akroyd et al. [8] and Vergara et al. [22] position federated digital twins as a way to establish semantic and interoperable connections among heterogeneous twins, enabling cooperation on complex tasks and supporting broader cross-domain perspectives. Yu et al. [10] extend this vision through the Internet of Federated Digital Twins, which aims to connect fragmented and heterogeneous twins into large-scale ecosystems capable of delivering services at societal scale. Kim et al. [11] address the computational heterogeneity of connected twins by proposing scheduling mechanisms that account for differences in processing capabilities among federated components.

Other works focus on more specific aspects of federation. Kavakli-Thorne et al. [13] study federated data exchange architectures in smart city digital twins. Memon et al. [12] address interoperability of AI-enhanced digital twins. Ricci et al. [23] and Giulianelli et al. [24] develop web-based and ecosystem-oriented approaches for interoperable digital twins, while Souza et al. [15] move toward federated, autonomous, and cognitive digital twins. These works show that federation is used to describe data exchange, interoperability, collaborative decision-making, platform-level coordination, and autonomous ecosystems. Gervasi et al. [18] further confirms this heterogeneity. In their classification, federated digital twins integrate independent and potentially autonomous twins through secure virtual-to-virtual interactions while preserving operational autonomy. The shared resources may include operational data, simulation outputs, knowledge artifacts, learning models, and agent-driven decisions. This observation directly motivates the federation-depth dimension proposed in our taxonomy: operational data correspond to digital-shadow federation, simulation and analytical outputs to descriptive or predictive model federation, learning models to model-level federation, and agent-driven decisions to prescriptive or control federation.

4.3. Research Gap and Motivation for a Taxonomy

The reviewed literature demonstrates growing interest in connecting multiple digital twins across systems, organizations, and domains. Existing works have introduced concepts such as systems of digital twins, digital twin networks, composite digital twins, federated digital twins, and digital twin ecosystems. These studies collectively highlight the importance of

interoperability, information exchange, distributed intelligence, and ecosystem-level coordination.

However, the notion of federation remains interpreted in substantially different ways. Some approaches view federation primarily as semantic interoperability and data exchange among heterogeneous twins. Others emphasize hierarchical composition of twins into higher-level representations, distributed deployment and execution, collaborative optimization, or autonomous ecosystem coordination. As a result, the same term is frequently used to describe fundamentally different architectural and interaction patterns.

Existing classifications primarily focus on structural forms of multi-twin systems, such as networks, aggregates, composites, federations, or ecosystems. In contrast, limited attention has been given to systematically characterizing how federation is realized within these structures. In particular, the literature lacks a unified framework for distinguishing the integration level of participating systems, the direction of federation, the abstraction levels involved, the depth of exchanged information, federation-aware behavior, runtime composition mechanisms, and interaction modes.

To address this gap, we develop a taxonomy of federated digital twins that focuses explicitly on federation semantics. Rather than classifying only the structural organization of multiple twins, the proposed taxonomy provides a systematic framework for describing how digital twins cooperate, exchange information, and collectively contribute to higher-level cyber-physical and systems-of-systems functionality.

5. Taxonomy Construction

To address the identified ambiguity, we constructed a taxonomy of federated digital twins intended to distinguish different architectural and interaction patterns in cyber-physical systems. The taxonomy was developed through conceptual decomposition and iterative refinement using the microgrid running example introduced in this section. The method consisted of four steps.

Step 1: Identification of recurring ambiguity. We first identified recurring ambiguities in the use of the term federation in digital twin and cyber-physical system contexts. These ambiguities concern what is federated, how twins are structurally related, whether a new higher-level twin is created, whether control remains local or is delegated, and whether participating twins are explicitly designed for federation. This step motivated the need to distinguish data, model, and control federation rather than treating federation as a binary property.

Step 2: Decomposition into conceptual dimensions. We then decomposed the design space into dimensions derived from three foundations: cyber-physical integration levels, digital twin abstraction levels, and federation concepts from distributed computing [25, 26] and systems-of-systems engineering. Cyber-physical integration levels distinguish modules, systems, and systems of systems. Digital twin abstraction levels distinguish digital shadows, descriptive models, predictive models, and prescriptive models [27]. Federation concepts

distinguish peer coordination, hierarchical composition, autonomy, runtime composition, and federation-aware interfaces. This decomposition resulted in seven taxonomy dimensions: integration level, federation direction, twin abstraction level, runtime composition, federation depth, federation awareness, and exchange mode.

Step 3: Construction of taxonomy categories. For each dimension, we defined a small set of categories that are intended to be mutually understandable and practically useful for engineering decisions. For example, federation direction distinguishes horizontal, vertical, and hybrid federation; federation depth distinguishes data/shadow-level, descriptive-model, predictive-model, and prescriptive/control-level federation; and federation awareness distinguishes full federation from para federation. The categories were selected to capture architectural differences that affect functionality, interoperability, autonomy, and control responsibility.

Step 4: Consistency check through a running example. Finally, we applied the taxonomy to a microgrid example composed of PV generation, battery storage, and EV charging systems. The example was used as a consistency check because it contains heterogeneous assets, local controllers, shared electrical coupling, and multiple possible federation configurations. In particular, the same assets can illustrate horizontal federation between battery and EV charger twins, vertical federation into a parent microgrid twin, and hybrid federation across multiple sites. This application helps demonstrate that the taxonomy can distinguish information sharing, distributed coordination, and higher-level twin creation.

The resulting taxonomy should be interpreted as a conceptual framework rather than a closed classification scheme. Its purpose is to support precise reasoning about federated digital twin architectures and to guide future implementation work, such as model-driven profiles, federation-aware interfaces, and deployment frameworks. The taxonomy is therefore evaluated in this paper by analytical clarity and applicability to the running example, while empirical evaluation of specific tooling and runtime performance is left for future work.

6. A Taxonomy of Federated Digital Twins

This section presents the proposed taxonomy of federated digital twins. Based on the foundations and observations introduced in the previous sections, we propose a taxonomy of federated digital twins organized into seven dimensions. The dimensions are grouped into two categories: architectural dimensions and interaction dimensions. The *architectural* dimensions describe the structural and organizational properties of federated digital twins. These include the integration level of the represented cyber-physical entities, the direction of federation among systems, the abstraction level of the participating twins, and the runtime composition of the federation architecture. The *interaction* dimensions describe how participating twins exchange information, models, predictions, and control decisions. These dimensions include the federation depth, the awareness of participating twins regarding federation, and the exchange mode used for synchronization and coordination. The

dimensions are intended to be orthogonal, but not necessarily independent; certain combinations naturally co-occur in practice. The same federation architecture may therefore combine different integration levels, abstraction levels, exchange modes, and federation depths depending on the participating systems and operational objectives. Table 1 summarizes the seven dimensions. The remainder of this section defines each dimension and explains how it contributes to the design space of federated digital twins.

The taxonomy is intended to support both analysis and engineering. As an analytical tool, it helps classify existing federated digital twin architectures and compare them along common dimensions. As an engineering tool, it helps designers decide which parts of a twin should be exposed, replicated, abstracted, composed, or delegated in a federation. This is particularly important for cyber-physical systems, where incorrect assumptions about control ownership, data freshness, or model authority may lead to inefficient or unsafe behavior. Therefore, the taxonomy explicitly separates information sharing, model-level reasoning, and control-level decision making.

6.1. Integration Level Dimension

As introduced in Section 2, digital twins may represent modules, systems, or systems of systems. In federated settings, this distinction is important because federation usually operates on exposed functionalities, capabilities, constraints, and control interfaces rather than on complete internal physical representations. Therefore, the *integration level* dimension does not only distinguish the scale of the represented entity, but also determines the abstraction boundary at which federation occurs.

At the *module level*, federation concerns individual sensing, actuation, communication, or control components. Examples include power meters, plug-detection sensors, inverters, battery-management controllers, or local optimization modules. Module-level federation is typically lightweight and tightly coupled to specific technical interfaces. It is useful when fine-grained synchronization or hardware-level coordination is required, but it usually exposes low-level operational information rather than system functionality.

At the *system level*, federation concerns complete functional systems composed of several modules. Examples include PV systems, stationary battery systems, EV chargers, building energy systems, or heat pumps. System-level twins are particularly relevant for federated digital twins because they expose meaningful operational functionality and constraints while abstracting away unnecessary internal implementation details. For example, an EV charger coordinating with a battery twin typically does not require detailed electrochemical models of lithium-ion cells or internal battery thermal dynamics. Instead, the federation may only expose operational properties such as state of charge, available charging or discharging power, response constraints, and safety limits. More detailed models may still exist locally for diagnostics, degradation estimation, or predictive maintenance, but they remain encapsulated within the local twin and are abstracted from the federation interface. This separation is important because it allows federation to fo-

Table 1: Seven dimensions of the proposed taxonomy organized into two groups: architectural and interaction dimensions.

Architectural Dimensions	
Integration level	Distinguishes whether twins represent modules, systems, or systems-of-systems.
Federation direction	Distinguishes horizontal, vertical, and hybrid federation structures among systems and twins.
Twin abstraction level	Captures whether twins expose digital shadows, descriptive, predictive, or prescriptive models.
Runtime composition	Captures whether federation structures are static or dynamic.
Interaction Dimensions	
Federation depth	Captures which parts of the twins are exchanged, shared, coordinated, or delegated.
Federation awareness	Distinguishes full federation from para federation based on federation-aware interaction capabilities.
Exchange mode	Captures how twins exchange information, including shared repositories, pull/push interaction, publish–subscribe communication, and negotiated exchange.

cus on interoperable capabilities rather than complete internal physical representations.

At the *system-of-systems* level, federation concerns independently managed systems coordinated toward shared or emergent objectives. Examples include microgrids, energy communities, industrial production lines, or aggregator portfolios. At this level, federation must additionally address autonomy, organizational boundaries, distributed decision making, and emergent functionality. For instance, several microgrid twins may coordinate flexibility exchange or grid-support services while preserving local optimization strategies and operational independence.

The integration level is therefore closely related to the amount of abstraction and encapsulation present in the federation. Lower integration levels expose more implementation-specific details, while higher integration levels expose system capabilities, flexibility, constraints, and operational objectives. Consequently, the same physical asset may participate in federation at different integration levels depending on the intended functionality. A battery may expose only module-level telemetry for diagnostics, system-level flexibility for local charging coordination, or system-of-systems level commitments within an aggregator federation.

6.2. Federation Direction Dimension

Federation direction defines how participating digital twins are structurally connected and coordinated within a federated architecture. The direction of federation determines whether twins interact as autonomous peers (horizontal direction), are composed into higher-level twins (vertical direction), or combine both coordination patterns (hybrid).

Horizontal federation refers to coordination among peer digital twins that preserve their operational autonomy and local control strategies. Participating twins exchange information, predictions, constraints, or control intentions to coordinate behavior without introducing a new parent twin responsible for global orchestration. Each twin continues to optimize and control its corresponding physical system independently while considering information received from other twins. Figure 5 illustrates a horizontal federation between a battery twin and an EV charger twin. The EV charger twin may expose charging demand, charging constraints, or expected charging duration,

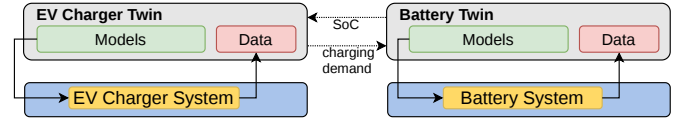


Figure 5: Horizontal federation between EV charger and battery twins through exchange of operational information creates a new functionality "battery supported EV charging", while preserving local autonomy and control ownership to the respective actual systems.

while the battery twin may expose state-of-charge information, available discharging flexibility, or charging/discharging constraints. Based on this exchanged information, the twins coordinate their behavior while preserving local decision-making logic. For example, the EV charger may reduce charging power or delay charging when the battery twin indicates insufficient available energy or discharging capability. Similarly, the battery twin may decide to discharge when charging demand is detected. The resulting federation enables safe battery-supported EV charging while preserving the autonomy of the participating twins.

Vertical federation refers to the composition of lower-level digital twins into a higher-level parent twin that introduces new system-level functionality and coordination objectives. In this case, participating twins become part of a larger federated digital twin that maintains a system-level view of the participating systems and may partially or fully coordinate their behavior. Vertical federation therefore introduces a new abstraction layer in which emergent functionality is realized at the federated level rather than solely within the individual twins. Figure 6 illustrates a vertical federation in which PV, battery, and EV charger twins are composed into a parent microgrid twin. While the individual twins continue to represent their corresponding physical systems, the parent twin introduces higher-level objectives such as maximizing self-consumption, coordinating flexibility, reducing grid import peaks, or optimizing energy flows across generation, storage, demand, and grid interaction. In contrast to horizontal federation, vertical federation may involve partial delegation of control decisions from child twins to the parent twin.

Hybrid federation combines horizontal and vertical federation within the same federated architecture. In such cases, local subsystems may internally rely on vertical federation while

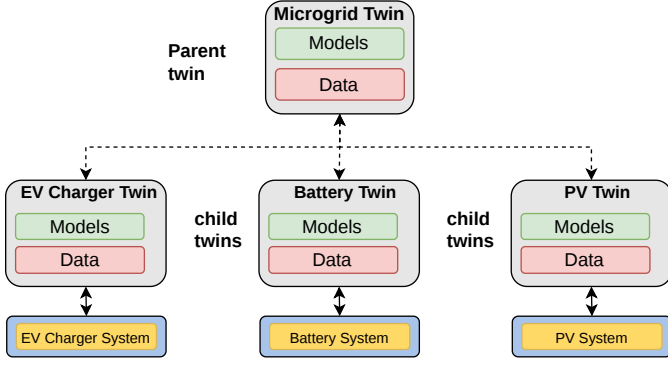


Figure 6: Vertical federation between the child (existing) twins: EV Charger, Battery, and PV Twins. A new parent Microgrid Twin is created, which introduces higher-level functionalities (objectives).

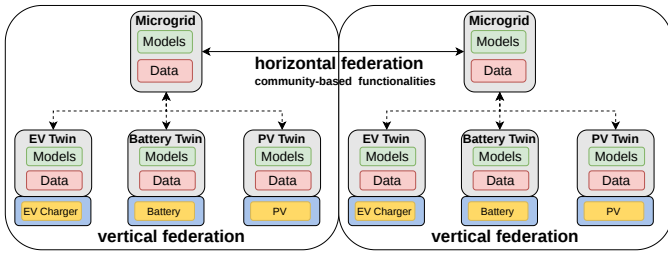


Figure 7: Hybrid federation: A microgrid twin is a parent twin for respective EV, battery, and PV child twins, but is also federated horizontally other microgrid twins to form energy community based functionalities.

simultaneously coordinating horizontally with other federated twins or organizational domains. Figure 7 illustrates a hybrid federation in which several vertically federated microgrid twins coordinate horizontally across sites or organizations. Each local microgrid twin internally orchestrates its own PV, battery, and EV charging systems, while higher-level coordination among microgrid twins enables energy exchange, distributed flexibility coordination, or cross-site optimization. Hybrid federation is particularly relevant in distributed energy communities, industrial ecosystems, and large-scale cyber-physical infrastructures where both local orchestration and inter-organizational coordination are required.

6.3. Twin Abstraction Level Dimension

The third dimension identifies which conceptual parts of a digital twin participate in federation. A digital twin is not only a data endpoint connected to a cyber-physical system. Depending on its maturity and purpose, it may include synchronized data representations, descriptive models of the current state, predictive models of future behavior, and prescriptive models that recommend or decide actions. Federation can therefore occur at different abstraction levels, ranging from the exchange of raw or processed data to the coordination of control decisions. We distinguish four *twin abstraction levels*: digital shadow (data-oriented), descriptive model (semantic-coordination), predictive model (anticipatory coordination), and prescriptive model (control coordination).

A *digital shadow* is the data-oriented representation of the observed cyber-physical system. It contains synchronized or

periodically updated measurements, events, configurations, and status information. In an energy system, examples include battery state of charge, PV power production, grid import, charging power, plug-in status, and inverter availability. Digital shadows provide the data basis for higher-level reasoning, but they do not by themselves explain, predict, or prescribe system behavior.

A *descriptive model* interprets the digital shadow and derives meaningful state information from it. For example, a battery digital shadow may report a state of charge of 72% and zero current power. A descriptive model may interpret this as “battery available for discharge up to 5 kW” or “battery idle and within safe operating range”. Similarly, an EV charger may expose plug-in status and current power, while its descriptive model identifies the current charging-session state, such as plugged, charging, paused, completed, or fault.

A *predictive model* estimates future states or behavior based on current data, historical data, external forecasts, or simulation. In a microgrid, predictive models may estimate PV production from weather forecasts, battery state-of-charge evolution under candidate schedules, or expected EV charging demand. Predictive models are particularly important for federated twins because local predictions may be insufficient when the behavior of one system depends on the future behavior of another system.

A *prescriptive model* recommends or decides actions. It may compute setpoints, schedules, policies, or constraints for the cyber-physical system. Examples include a battery charge/discharge setpoint, an electric vehicle charging-power limit, or a PV curtailment command. Prescriptive models are closely related to control ownership. In a federated digital twin architecture, it is therefore important to specify whether prescriptive decisions remain local to child twins, are coordinated among peer twins, or are delegated to a higher-level federated twin.

This abstraction-level dimension is central to distinguishing different forms of federation. A federation that only exchanges digital shadows remains data-oriented. A federation that exchanges descriptive states enables semantic coordination. A federation that exchanges predictive models or forecasts enables anticipatory coordination. A federation that exchanges prescriptive decisions, constraints, or setpoints affects control behavior and therefore requires explicit assumptions about authority, safety, and responsibility.

The abstraction level also clarifies what may be exposed, replicated, abstracted, or delegated in a federation. In horizontal federation, peer twins may expose selected digital shadows or descriptive states while retaining their own predictive and prescriptive models. In vertical federation, child twins may expose data, descriptive states, predictions, or constraints to a parent federated twin, which constructs its own higher-level descriptive, predictive, and prescriptive models. However, even in such cases, the execution interface to the physical system usually remains with the child twins, because they maintain the direct connection to sensors, actuators, and local safety mechanisms.

6.4. Runtime Composition Dimension

Runtime composition defines whether participating digital twins and federation relationships are predefined before deployment or dynamically established during runtime operation. This dimension therefore captures how federated digital twins are discovered, bound, and composed into a federation architecture.

Static composition refers to federation architectures in which participating twins, interfaces, and coordination relationships are predefined before runtime execution. The federation structure is engineered and configured in advance, and the participating twins are typically known prior to deployment. Such architectures are common in industrial automation, microgrids, manufacturing systems, and engineered cyber-physical infrastructures where system boundaries and interactions remain relatively stable over time.

Dynamic composition refers to federation architectures in which participating twins may be discovered, added, removed, or rebound during runtime operation. In such cases, federation participants are not necessarily known in advance and may be identified through discovery services, negotiation protocols, capability matching, or decentralized coordination mechanisms. Dynamic composition is particularly relevant in large-scale distributed infrastructures where federation participants evolve over time or belong to different organizations and administrative domains. For example, EV charging systems, distributed storage assets, or building twins may become temporarily available and participate in local energy coordination based on operational conditions, discovered capabilities, or flexibility requirements. In such scenarios, the federation architecture must support runtime discovery, registration, negotiation, and adaptive coordination.

Static composition typically simplifies engineering, verification, and orchestration because federation participants and interfaces are known in advance. In contrast, dynamic composition improves flexibility, scalability, and adaptability, but introduces additional challenges related to discovery, compatibility, synchronization, trust management, and runtime coordination.

6.5. Federation Depth Dimension

Federation depth defines which parts of participating digital twins are exchanged, coordinated, replicated, or delegated within a federation architecture. Federation depth is closely related to the abstraction levels of digital twins introduced in Section 6.3, but the two dimensions are not equivalent. Twin abstraction level describes which functional layers are internally present within a digital twin, whereas federation depth describes which of these layers are shared or coordinated across participating twins. A twin may internally maintain predictive and prescriptive models while participating only in data-level federation. The proposed taxonomy distinguishes four federation depths: digital-shadow federation, descriptive-model federation, predictive-model federation, and prescriptive-model federation, as presented in Fig.8.

Digital-shadow federation represents the shallowest form of federation and involves the exchange or synchronization

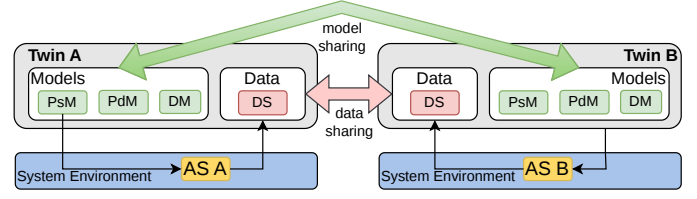


Figure 8: Federation depth: shadow-, descriptive-, predictive-, or prescriptive-model federation. While the models may be federated, the control to the physical system is retained by the original twin.

of operational measurements, events, configurations, or states maintained within digital shadows. Participating twins expose data without sharing higher-level interpretations, predictions, or control decisions. Such federation is commonly used for monitoring, visualization, state synchronization, and cross-system observability. A digital-shadow federation between battery and EV charger twins may be implemented as follows. The EV charger twin may expose charging demand, charging power, or plug state, while the battery twin may expose state of charge, available power, or operational availability. The participating twins exchange synchronized operational information while preserving independent descriptive, predictive, and prescriptive logic.

Descriptive-model federation extends data federation by sharing interpreted operational states, semantic descriptions, or derived contextual information. Instead of exchanging only raw measurements, participating twins expose higher-level interpretations of their current operational conditions. For example, a battery twin may expose whether charging or discharging is currently possible, while an EV charger twin may expose charging-session status, estimated availability, or operational readiness.

Predictive-model federation additionally exchanges future behavior estimations, forecasts, or predictive capabilities among participating twins. Participating twins may share predicted energy demand, generation forecasts, expected charging completion time, battery availability forecasts, or future operational constraints. Such federation enables anticipatory coordination and predictive optimization across multiple systems.

Prescriptive-model federation represents the deepest level of federation and involves the exchange, coordination, or partial delegation of decisions, optimization objectives, constraints, or control strategies. At this level, federation affects not only monitoring and prediction, but also operational decision making and control ownership. Participating twins may expose control interfaces, optimization objectives, flexibility constraints, or executable strategies to other twins or to a higher-level federated twin.

The federation depths are cumulative. Higher federation depths typically depend on lower-level synchronization and information exchange. For example, predictive-model federation generally requires synchronized operational data and descriptive operational states, while prescriptive-model federation often depends on predictive coordination among participating twins.

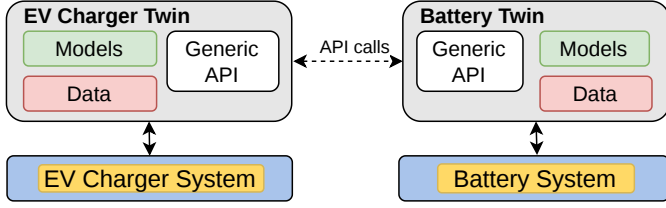


Figure 9: Full federation: Exchange e.g., plug state, charging demand, or SoC.

6.6. Federation Awareness Dimension

Federation awareness defines whether participating twins are explicitly designed to support federation-specific coordination capabilities or are integrated through generic interfaces without knowledge of federation semantics. This dimension therefore captures the degree to which federation is visible and explicitly represented within the participating twins themselves. The proposed taxonomy distinguishes two forms of federation awareness: full federation and para federation.

Full federation refers to federation architectures in which participating twins expose generic monitoring, prediction, or control interfaces without explicitly embedding federation-aware functionality. In such cases, the twins themselves are not aware that they participate in a federation. Instead, federation logic is externally imposed through orchestration, integration middleware, shared repositories, or parent twins that interpret and coordinate the exposed interfaces. Figure 9 illustrates full federation between battery and EV charger twins. The participating twins expose operational information such as charging demand, state-of-charge values, charging schedules, or operational constraints through generic interfaces. However, the twins do not explicitly expose federation-specific capabilities, negotiation strategies, coordination semantics, or knowledge about other twins. The federation is therefore realized as a black-box integration of independently developed twins. Full federation simplifies interoperability because existing digital twins can participate in federation without requiring internal redesign. This is particularly useful for integrating legacy twins, third-party systems, or independently operated platforms. However, federation coordination remains external to the twins themselves, which may limit adaptability, negotiation capabilities, or semantic coordination.

Para federation refers to federation architectures in which participating twins explicitly expose federation-aware capabilities, constraints, coordination semantics, or negotiation interfaces. In such cases, the twins are aware that they participate in a federation and internally maintain federation-related logic, metadata, or interaction capabilities. Figure 10 illustrates para federation in the same federation scenario of EV charger and battery twins. Participating twins may expose not only operational states and predictions, but also flexibility descriptions, negotiation policies, optimization objectives, capability advertisements, federation constraints, trust relationships, or semantic coordination interfaces. For example, a battery twin may advertise flexibility windows, degradation-aware discharge constraints, or negotiation policies for energy exchange, while an EV charger twin may expose charging priorities, flexibility

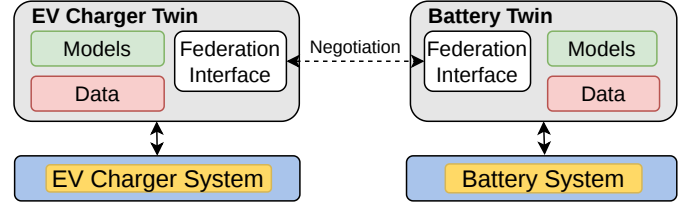


Figure 10: Para federation: Expose not only plug state or SoC, but also e.g., flexibility descriptions, optimization objectives, or federation constraints.

preferences, or reservation constraints. Para federation enables richer decentralized coordination, runtime negotiation, adaptive orchestration, and autonomous decision making among participating twins. Such capabilities are particularly relevant in dynamic energy communities, cross-organizational infrastructures, decentralized flexibility markets, and large-scale cyber-physical ecosystems where federation relationships evolve during runtime operation.

6.7. Federation Exchange Mode

The previous dimensions define what is federated and which parts of the twins participate in federation. However, federated digital twins may also differ in how data, models, predictions, constraints, or control decisions are exchanged. We refer to this dimension as *exchange mode*. It captures the interaction mechanism used among participating twins and therefore affects latency, coupling, synchronization, scalability, autonomy, and runtime coordination.

We distinguish five representative exchange modes: shared repository exchange, pull-based exchange, push-based exchange, publish-subscribe exchange, and negotiated exchange, as summarized in Fig. 11. These modes are not mutually exclusive and may coexist within the same federation architecture.

In *shared repository exchange* (Fig. 11 a)), participating twins access a common storage or shared data space. The shared repository may contain synchronized digital shadows, semantic states, forecasts, historical data, constraints, or optimization results. This mode simplifies interoperability because twins do not need direct peer-to-peer communication. For example, several energy twins may access a shared microgrid repository containing PV production, battery state of charge, charging-session states, and grid measurements. A parent microgrid twin may aggregate this information into a federated shadow and optimization model. Shared repositories reduce communication complexity but increase dependence on centralized storage and synchronization mechanisms. They are therefore common in vertically federated architectures and cloud-based digital twin platforms.

In *pull-based exchange* (Fig. 11 b)), one twin explicitly requests information from another twin when needed. The requesting twin controls when synchronization occurs and which information is retrieved. For example, an EV charger twin may periodically query a battery twin to retrieve the current state of charge and available discharge power before deciding whether charging can start or continue safely. Pull-based exchange reduces unnecessary communication and allows twins to remain

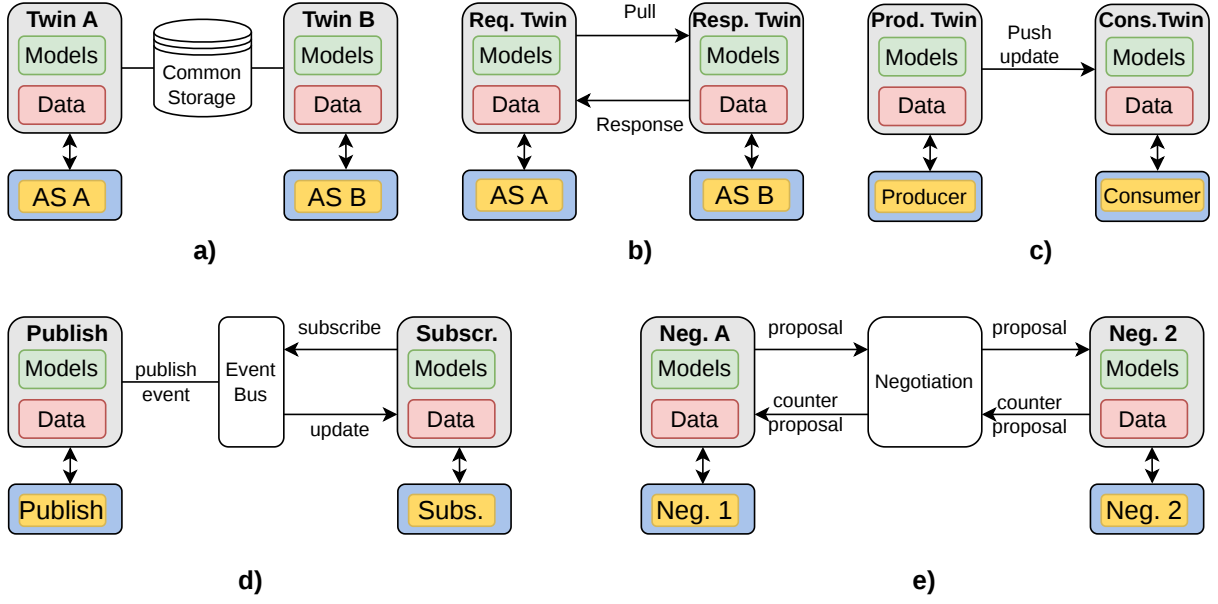


Figure 11: Exchange modes in federated digital twins.

relatively autonomous. However, it may increase latency or create stale information if polling intervals are too long.

In *push-based exchange* (Fig. 11 c)), a twin proactively transmits updates when relevant changes occur. The producer twin controls when information is disseminated. For instance, a battery twin may immediately notify a charging twin when the available discharge power drops below a safety threshold or when the battery becomes unavailable for discharging. Push-based exchange improves responsiveness and reduces the need for polling, but it may increase communication frequency and coupling among participating twins.

In *publish-subscribe exchange* (Fig. 11 d)), twins communicate through event-driven messaging infrastructures. Producer twins publish events or state updates to topics, while consumer twins subscribe to the topics relevant to their operation. For example, PV, battery, and EV charger twins may publish production, flexibility, or charging events to a microgrid event bus. A parent microgrid twin or peer twins can subscribe dynamically to the required streams. This mode supports scalable and loosely coupled federation, particularly in dynamic or system-of-systems environments. Publish-subscribe exchange is especially suitable for hybrid and dynamic federation because new twins may join or leave the federation without requiring direct reconfiguration of existing twins.

In *negotiated exchange* (Fig. 11 e)), twins exchange not only data or state information, but also capabilities, constraints, preferences, and proposed actions. The participating twins may iteratively negotiate coordination decisions before execution. For example, a battery twin may expose a flexibility envelope and degradation constraints, while an EV charger twin exposes charging urgency and maximum acceptable charging power. The twins may negotiate a mutually acceptable charging plan instead of directly exchanging fixed setpoints. Negotiated exchange is closely related to para federation because it requires

federation-aware interfaces capable of describing operational constraints and coordination semantics. It is particularly relevant for deep predictive or prescriptive federation involving distributed optimization and decentralized decision making.

Exchange mode is orthogonal to the remaining taxonomy dimensions. For example, horizontal federation may use pull-based or negotiated exchange among peer twins, whereas vertical federation may rely on shared repositories or publish-subscribe infrastructures. Similarly, predictive or control-level federation may use either centralized or event-driven exchange mechanisms depending on latency, autonomy, and scalability requirements.

6.8. Cross-Dimensional View

The seven dimensions introduced above are intended to be used jointly rather than independently. A federated digital twin architecture is characterized not only by whether it is horizontal or vertical, but also by the twin abstraction levels involved, the integration level at which federation occurs, the runtime composition model, the depth of federation, the degree of federation awareness, and the exchange mode. This cross-dimensional view helps avoid ambiguous statements such as “the twins are federated” without specifying what is actually federated and how.

Table 2 illustrates representative combinations of the taxonomy dimensions. The examples are not exhaustive, but they show how different architectural choices lead to different forms of federation. The table highlights three important observations. *First*, federation direction alone is insufficient to characterize an architecture. A horizontal federation may be a simple monitoring federation if peer twins only exchange digital shadows, or it may become a distributed-control federation if peer twins exchange constraints and adapt local decisions. Similarly, a vertical federation may merely aggregate data from child

Table 2: Representative cross-dimensional configurations of federated digital twins.

Configuration	Integration level	Direction	Depth	Awareness	Runtime composition
Monitoring federation	System	Horizontal	Data/shadow	Full	Static or dynamic
Semantic coordination federation	System	Horizontal	Descriptive-model	Full or para	Static or dynamic
Predictive coordination federation	System or system-of-systems	Horizontal	Predictive-model	Para	Dynamic
Distributed-control federation	System	Horizontal	Prescriptive/control	Para	Dynamic
Parent-child aggregation federation	System	Vertical	Data/shadow or descriptive-model	Full	Static
Vertical optimization federation	System to system-of-systems	Vertical	Predictive and prescriptive/control	Full or para	Static or dynamic
Capability-aware vertical federation	System to system-of-systems	Vertical	Descriptive, predictive, and prescriptive/control	Para	Static or dynamic
Multi-site hybrid federation	System-of-systems	Hybrid	Data, model, and control	Para	Dynamic
Microgrid (PV–Battery–EV) DT	System to system-of-systems	Vertical	Shadow + Prescriptive	Full	Static

twins, or it may create a higher-level optimization twin with predictive and prescriptive models. *Second*, federation depth and federation awareness are closely related. Deep federation becomes more robust when participating twins are federation-aware. For example, prescriptive/control-level federation is possible with generic interfaces, but it is safer and more explainable when child twins expose constraints, flexibility envelopes, and local safety conditions. This is the main motivation for para federation. *Third*, runtime composition determines whether the federation can adapt to changing cyber-physical and organizational contexts. Static vertical federation may be sufficient for a fixed microgrid with known assets. Dynamic or system-of-systems federation is required when assets join and leave, when multiple organizations coordinate, or when the federation is created only under specific operational conditions.

Based on the taxonomy, we can formulate a more precise definition of federated digital twins:

A federated digital twin is a structured composition or coordination of multiple digital twins in which selected data, models, decisions, or capabilities are exposed, exchanged, composed, or delegated across twin boundaries, while preserving explicit relationships to the underlying cyber-physical systems and their control responsibilities.

This definition intentionally includes both horizontal and vertical federation. In horizontal federation, the composition is primarily coordination among peer twins. In vertical federation, the composition creates a higher-level parent twin. In hybrid federation, both forms coexist. The definition also emphasizes control responsibilities, because in cyber-physical systems the question of who observes, predicts, decides, validates, and

executes is as important as the question of which data are exchanged.

The cross-dimensional view provides a basis for the application example in the next section. The microgrid example demonstrates how the same physical systems can participate in different federation configurations: a battery and EV charger can form a horizontal distributed-control federation, while photovoltaic, battery, and EV charger twins can be vertically composed into a microgrid twin with system-level optimization functionality.

7. Prototype Realization of a Vertically Federated Microgrid Twin

For the purpose of this paper, we developed a prototype model-driven library for constructing federated digital twins from SysML v2¹ specifications. The library is intentionally designed to be use-case agnostic and therefore does not prescribe domain-specific analytical models. Instead, domain experts define executable digital-twin behavior externally in the form of serverless functions (AWS Lambda) and workflows (AWS Step Functions) associated with the digital-twin models.

To demonstrate the applicability of the proposed taxonomy and federation concepts, we developed a prototype realization of a vertically federated microgrid digital twin, using the prototype library. The prototype integrates three system-level twins representing a PV system, a battery storage system, and an EV charging infrastructure into a higher-level microgrid twin. The implementation follows a vertical federation approach, in

¹<https://www.omg.org/sysml/sysmlv2/>

which the parent microgrid twin combines information and control capabilities from the child twins in order to realize system-level energy-management functionality.

In the current prototype, the primary focus is on the federation of digital shadows and prescriptive models. The prescriptive logic is implemented through executable control strategies associated with the participating twins. As an initial proof of concept, we implemented a battery charging and discharging strategy that uses information obtained from the other twins, particularly the aggregated net demand within the microgrid. The prototype therefore demonstrates how vertically federated twins can combine system-level data and control logic into a coordinated operational strategy while preserving the modularity of the participating twins.

7.1. Prototype Modeling Library

The prototype is realized using a model-driven library for describing federated digital twins using SysML v2-based specifications. The goal of the library is not to define domain-specific digital-twin semantics, but rather to provide a generic modeling structure that can be specialized for different cyber-physical systems and federation scenarios.

The library extends SysML v2 with lightweight constructs for describing digital twins, entities, components, telemetry attributes, executable strategies, and federation-related relationships. The modeling approach distinguishes between type-level definitions and deployable instances. Type-level definitions describe reusable digital-twin structures and component abstractions, while instance-level elements represent concrete digital-twin realizations participating in a federation.

The library supports hierarchical entity structures in which a digital twin may contain multiple entities and components. Entities represent logical or physical assets such as batteries, EV chargers, PV systems, or microgrids, while components represent functional units exposing telemetry and operational behavior. Telemetry values are represented through measurement attributes associated with components.

Executable digital-twin behavior is represented through conditions and strategy actions. Conditions define logical relations over telemetry values and configuration attributes, while strategy actions define executable operations associated with the digital twin. The current prototype uses these strategy actions to represent prescriptive control logic within the federated microgrid twin.

The library intentionally remains agnostic regarding the implementation of descriptive, predictive, or prescriptive models. Instead of prescribing analytical methods, the library allows domain experts to externally define executable logic associated with the digital twins. This enables the integration of heterogeneous domain-specific models and workflows without constraining the implementation to a predefined analytical approach. The prototype currently supports the representation of: hierarchical digital-twin structures, telemetry attributes and synchronized digital shadows, executable strategy actions, feedback relations among twins and components, and federated parent-child twin relationships.

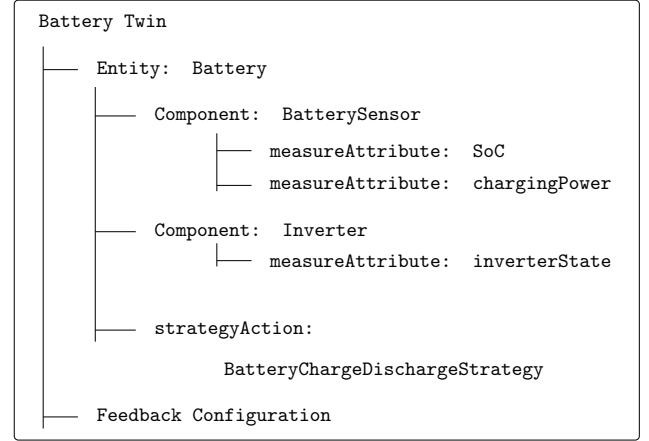


Figure 12: Battery twin implementation with the prototype library

The resulting SysML v2 specifications are transformed into JSON-based deployment and execution configurations. These configurations describe the hierarchical organization of twins, telemetry mappings, executable strategy definitions, and feedback relations among participating twins.

7.2. Modeling Individual System-Level Twins

Using the prototype library, we modeled three system-level digital twins representing a PV system, a battery storage system, and an EV charging infrastructure. Each twin is represented as a hierarchical SysML v2 specification containing entities, components, telemetry attributes, and executable strategy actions.

The battery twin represents the operational state and control behavior of the battery storage system. The model includes telemetry attributes such as state of charge and charging and discharging power, as presented in Fig. 12. In addition to the synchronized digital shadow, the battery twin contains prescriptive strategy actions responsible for charging and discharging decisions. In the current prototype, this prescriptive logic forms the core operational strategy of the federated microgrid twin. The implemented battery strategy uses information obtained from the federated twins, particularly aggregated net demand within the microgrid. Based on this information, the strategy determines whether the battery should charge, discharge, or remain idle. The strategy therefore represents an initial realization of prescriptive-model federation in which system-level information influences the operational behavior of individual twins.

Listing 1 presents a simplified JSON-based representation of the battery digital twin generated from the SysML v2 specification. The representation captures the hierarchical organization of the twin, including entities, components, telemetry attributes representing the digital shadow, and executable strategy actions implementing prescriptive behavior. The example additionally illustrates how executable prescriptive logic is associated with the digital twin through strategy actions. In the current prototype, such strategy actions implement battery charging and discharging behavior based on system-level information obtained through the federated microgrid twin.

Listing 1: Simplified representation of the battery twin structure.

```

1 {
2   "twin": "BatteryTwin",
3   "entity": "Battery",
4   "components": [
5     {
6       "name": "BatterySensor",
7       "measureAttributes": [
8         "stateOfCharge",
9         "chargingPower"
10      ]
11    }
12  ],
13  "strategyActions": [
14    "BatteryChargeDischargeStrategy"
15  ]
16 }

```

The EV charging twin represents the operational state and charging behavior of the EV charging infrastructure. The model includes telemetry attributes describing charging demand, charging power, and charging-related operational states. Within the prototype library, these telemetry values are represented through measurement attributes associated with charging components. The EV charging twin additionally exposes executable strategy interfaces that allow charging behavior to be coordinated by higher-level federation logic.

The PV twin represents the energy-generation subsystem of the microgrid. The model includes telemetry attributes describing generated power, inverter-related operational states, and energy-production information. The PV twin primarily contributes synchronized operational information to the federation and participates in the calculation of aggregated microgrid demand and available local generation.

All three twins are modeled independently and preserve their own local digital-shadow representations and executable strategy definitions. The current prototype therefore follows a modular twin-engineering approach in which system-level coordination is introduced through federation rather than through monolithic digital-twin models.

The resulting SysML v2 specifications are transformed into JSON-based representations describing hierarchical entity structures, telemetry mappings, executable strategies, and feedback configurations. These representations are later used to construct the vertically federated microgrid twin described in the following subsection.

7.3. Constructing the Federated Microgrid Twin

The individual *system-level twins*² are *vertically federated* into a higher-level microgrid twin representing the coordinated operation of the PV system, battery storage system, and EV charging infrastructure. The resulting federated twin introduces

²We format the selected dimensions with italic in this section, which are summarized in Table 3 in the next Section 7.4.

system-level operational objectives and combines information obtained from the child twins into a unified control strategy. The federation architecture consists of a parent microgrid twin and three child twins representing the PV, battery, and EV charging systems. Each child twin preserves its own digital shadow and local executable strategies, while the parent twin introduces additional coordination logic operating at the microgrid level.

The federation primarily combines *digital-shadow* federation and *prescriptive-model federation*. At the digital-shadow level, the parent twin reads operational information from the participating twins, including PV generation, battery state-of-charge information, charging demand, and operational states. This information is used to derive aggregated microgrid-level operational conditions such as net demand and available local flexibility.

At the *prescriptive-model level*, the parent microgrid twin coordinates executable strategy actions associated with the participating twins. In the current implementation, the microgrid twin aggregates information from the child twins and determines charging and discharging behavior for the battery system based on system-level operational conditions. The resulting control strategy therefore represents a vertically federated prescriptive model in which local operational behavior is influenced by higher-level federation logic.

The current prototype follows a *static federation* approach in which the participating twins and their federation relations are known in advance. The parent twin therefore maintains predefined references to the child twins and directly accesses their synchronized telemetry information and executable strategy interfaces.

The current implementation follows a *full federation* approach. The child twins are not explicitly aware that they participate in a federation and do not expose federation-specific negotiation or coordination semantics. Instead, the parent microgrid twin accesses the exposed telemetry and strategy interfaces of the child twins through predefined relations and externally orchestrates the federation logic.

The federation currently uses a *pull-based exchange mode*. The parent microgrid twin periodically reads telemetry and operational information from the child twins in order to construct a system-level operational view of the microgrid. The child twins therefore preserve their own local state and execution logic while exposing synchronized operational data to the parent federation layer.

The federated twin is represented through JSON-based configurations generated from the SysML v2 specifications. These configurations describe the hierarchical federation structure, telemetry mappings, executable strategy actions, and feedback relations among the participating twins. The prototype currently includes strategy compositions such as combined PV–battery coordination and aggregated microgrid-level strategies integrating all participating twins.

Overall, the prototype demonstrates how the proposed taxonomy dimensions can be realized within a concrete federated digital-twin architecture. The implementation specifically illustrates vertical federation, static composition, digital-shadow

Table 3: Mapping of the prototype realization to the proposed taxonomy.

Taxonomy	Dimen- sion	Prototype Realization
Integration level		System-level twins federated into a microgrid twin
Federation direction		Vertical federation
Twin abstraction level		Digital shadows and executable prescriptive strategies
Runtime composition		Static composition
Federation depth		Digital-shadow and prescriptive-model federation
Federation awareness		Full federation
Exchange mode		Pull-based federation

and prescriptive-model federation depth, full federation awareness, and pull-based federation exchange.

7.4. Mapping the Prototype to the Taxonomy

Table 3 summarizes how the implemented prototype maps to the proposed taxonomy of federated digital twins. The prototype demonstrates how the proposed taxonomy dimensions can be instantiated within a concrete federated digital-twin realization. The implemented federation combines system-level twins into a higher-level microgrid twin while preserving modular twin structures and local operational representations. The prototype additionally illustrates that the taxonomy dimensions remain orthogonal, as federation direction, federation depth, runtime composition, federation awareness, and exchange mode are independently selected within the implemented architecture.

8. Discussion

The proposed taxonomy introduces a structured perspective on federated digital twins for cyber-physical systems-of-systems by separating architectural and interaction-related federation dimensions. In contrast to many existing discussions that treat federation primarily as data exchange or distributed deployment, our taxonomy explicitly distinguishes integration level, federation direction, twin abstraction level, runtime composition, federation depth, federation awareness, and exchange mode. The dimensions are intended to be orthogonal rather than mutually exclusive architectures. Consequently, different federation realizations may combine the dimensions in different ways depending on the operational objectives and engineering constraints of the target cyber-physical system.

One of the key observations emerging from the taxonomy is the explicit relationship between digital-twin abstraction levels and federation semantics. Existing digital-twin architectures often distinguish digital shadows, descriptive models, predictive models, and prescriptive models within individual twins, but they rarely distinguish which of these layers participate in federation. The proposed notion of federation depth therefore introduces a mechanism for describing how federation affects

different functional layers of the participating twins. This distinction becomes particularly important in large-scale cyber-physical systems where data sharing, predictive coordination, and control delegation may have substantially different operational and organizational implications.

The taxonomy additionally distinguishes federation direction from integration level. Vertical federation does not merely represent system hierarchy, but rather the creation of higher-level twins introducing new system-level functionality and coordination objectives. Similarly, horizontal federation captures peer coordination among autonomous twins without necessarily introducing centralized orchestration. This distinction enables a more precise characterization of federated digital-twin architectures across industrial, energy-management, and cyber-physical domains.

Another important contribution is the distinction between full and para federation awareness. Existing digital-twin systems frequently realize full federation in which independently developed twins expose generic interfaces while federation logic remains external to the twins themselves. However, future decentralized cyber-physical ecosystems are likely to require federation-aware twins capable of exposing semantic constraints, negotiation policies, flexibility descriptions, and runtime coordination capabilities. Para federation therefore provides an initial conceptual step toward autonomous and adaptive federation-aware digital-twin ecosystems.

The prototype realization demonstrates the applicability of the proposed taxonomy within a concrete vertically federated microgrid scenario. The implementation shows how system-level twins can be combined into a higher-level federated twin through digital-shadow and prescriptive-model federation. The prototype additionally illustrates how executable prescriptive logic can be represented through strategy actions associated with the twins. Finally, the taxonomy currently focuses on technical federation semantics and does not explicitly model governance, ownership, trust boundaries, or contractual federation agreements between organizations.

Nevertheless, the current prototype represents only an initial realization of the proposed concepts. The implementation focuses on static vertical federation and primarily realizes digital-shadow and prescriptive-model federation. Dynamic runtime composition, predictive-model federation, negotiation-based exchange, and para federation are currently not fully implemented and remain part of future work. Similarly, the current prototype focuses on a single microgrid scenario and does not yet evaluate scalability, interoperability, or cross-domain federation behavior. Furthermore, the current taxonomy primarily focuses on structural and semantic federation properties. Although runtime composition distinguishes static and dynamic federation, fully decentralized runtime federation involving autonomous discovery, negotiation, and temporary federation contracts remains an open research challenge. Finally, the taxonomy was developed through conceptual synthesis and has currently been evaluated on a single microgrid case study.

Future work will therefore focus on extending the prototype toward dynamic federation mechanisms, federation-aware twins, predictive-model coordination, and larger-scale multi-

domain federation scenarios. In particular, future research will investigate runtime twin discovery, negotiation-based federation, distributed coordination among energy communities, and engineering support for federated digital-twin development and deployment.

A further extension concerns service- or capability-based federation. The current taxonomy distinguishes how twins are organized, what layers participate in federation, and how information is exchanged. However, dynamic cyber-physical ecosystems may also require explicit descriptions of the services or capabilities for which federation is established. For example, an EV twin may discover an EV charger twin through a physical plug event, but the federation still needs to bind a specific capability such as charging, bidirectional energy exchange, authentication, billing, or flexibility provision. Future work will therefore also investigate whether service semantics, capability discovery, and contract-based binding should form an additional taxonomy dimension for dynamic and cross-domain federated digital twins.

9. Conclusion

This paper presented a taxonomy of federated digital twins for cyber-physical and industrial systems. The taxonomy distinguishes seven dimensions organized into architectural and interaction dimensions, including integration level, federation direction, twin abstraction level, runtime composition, federation depth, federation awareness, and exchange mode. The proposed taxonomy addresses the current lack of structured concepts for describing how digital twins exchange information, coordinate behavior, and construct higher-level federated functionality.

A key contribution of the taxonomy is the explicit distinction between the internal abstraction layers of digital twins and the federation depth through which these layers participate in federation. The taxonomy additionally distinguishes horizontal, vertical, and hybrid federation structures while introducing the notions of full and para federation awareness. Together, these dimensions provide a conceptual foundation for describing federated digital-twin architectures across energy systems, industrial infrastructures, and large-scale cyber-physical ecosystems.

To demonstrate the applicability of the taxonomy, the paper presented a prototype realization of a vertically federated microgrid twin integrating PV, battery, and EV charging system-level twins. The prototype illustrates how digital-shadow and prescriptive-model federation can be realized through executable strategy actions and parent-child federation structures.

Overall, the proposed taxonomy provides a foundation for future research on federated digital twins, including dynamic runtime federation, federation-aware coordination, negotiation-based exchange, predictive-model federation, and large-scale decentralized cyber-physical ecosystems.

Acknowledgment

This work was funded by *CETPartnership, the Clean Energy Transition Partnership* under the 2024 joint call for research

proposals (the **FLEXI** project), co-funded by the European Commission (GA N°101069750) and with the funding organizations FFG Austrian Research Promotion Agency (Austria), NWO Dutch Research Council (the Netherlands), Swedish Energy Agency (Sweden) and GSRI (Greece); the Key Digital Technologies (KDT) Joint Undertaking through the project **MATISSE**, grant agreement No. 101140216; and *Land Tirol, Austria*, under contract F.35499 (the **TIM** project).

The authors used generative AI (OpenAI GPT-5) for grammar, language editing, and formatting the tables in Latex. All content was reviewed and approved by the authors.

References

- [1] W. Kritzing, M. Karner, G. Traar, J. Henjes, W. Sihn, Digital twin in manufacturing: A categorical literature review and classification, *IFAC-PapersOnLine* 51 (11) (2018) 1016–1022, 16th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2018. doi:<https://doi.org/10.1016/j.ifacol.2018.08.474>.
- [2] R. Eramo, F. Bordeleau, B. Combemale, M. v. d. Brand, M. Wimmer, A. Wortmann, Conceptualizing digital twins, *IEEE Software* 39 (2) (2022) 39–46. doi:[10.1109/MS.2021.3130755](https://doi.org/10.1109/MS.2021.3130755).
- [3] F. Tao, H. Zhang, A. Liu, A. Y. C. Nee, Digital twin in industry: State-of-the-art, *IEEE Transactions on Industrial Informatics* 15 (4) (2019) 2405–2415. doi:[10.1109/TII.2018.2873186](https://doi.org/10.1109/TII.2018.2873186).
- [4] Q. Lu, A. K. Parlikad, P. Woodall, G. D. Ranasinghe, X. Xie, Z. Liang, E. Konstantinou, J. Heaton, J. Schooling, Developing a digital twin at building and city levels: Case study of west cambridge campus, *Journal of Management in Engineering* 36 (3) (2020) 05020004. doi:[10.1061/\(ASCE\)ME.1943-5479.0000763](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000763).
- [5] R. Minerva, G. M. Lee, N. Crespi, Digital twin in the iot context: A survey on technical features, scenarios, and architectural models, *Proceedings of the IEEE* 108 (10) (2020) 1785–1824. doi:[10.1109/JPROC.2020.2998530](https://doi.org/10.1109/JPROC.2020.2998530).
- [6] E. Ferko, A. Bucaioni, M. Behnam, Architecting digital twins, *IEEE Access* 10 (2022) 50335–50350. doi:[10.1109/ACCESS.2022.3172964](https://doi.org/10.1109/ACCESS.2022.3172964).
- [7] Aeneas, EPoSS, INSIDE, Electronic components and systems: Strategic research and innovation agenda (2026).
- [8] J. Akroyd, S. Mosbach, A. Bhave, M. Kraft, Universal digital twin - a dynamic knowledge graph, *Data-Centric Engineering* 2 (2021) e14. doi:[10.1017/dce.2021.10](https://doi.org/10.1017/dce.2021.10).
- [9] C. Vergara, R. Bahsoon, G. Theodoropoulos, W. Yanez, N. Tziritas, Federated digital twin, in: 2023 IEEE/ACM 27th International Symposium on Distributed Simulation

- and Real Time Applications (DS-RT), 2023, pp. 115–116. doi:10.1109/DS-RT58998.2023.00027.
- [10] T. Yu, Z. Li, O. Hashash, K. Sakaguchi, W. Saad, M. Debah, Internet of federated digital twins: Connecting twins beyond borders for society 5.0, *IEEE Internet of Things Magazine* 7 (5) (2024) 64–71. doi:10.1109/IOTM.001.2300279.
- [11] Y.-J. Kim, H. Kim, B. Ha, W.-T. Kim, Federated digital twins: A scheduling approach based on temporal graph neural network and deep reinforcement learning, *IEEE Access* 13 (2025) 20763–20777. doi:10.1109/ACCESS.2025.3530558.
- [12] U. Memon, W. Mayer, M. Selway, M. Stumptner, Interoperability of ai-enhanced digital twins, *Journal of Industrial Information Integration* 48 (2025) 100961. doi:https://doi.org/10.1016/j.jii.2025.100961.
- [13] M. Kavakli-Thorne, J. Zarean, System architectures for data exchange in a regional federated digital twin: A case study on birmingham smart city, in: 2025 International Conference on Intelligent Computing and Virtual & Augmented Reality Simulations (ICVARS), 2025, pp. 43–49. doi:10.1109/ICVARS66454.2025.11198668.
- [14] P. Gritsch, D. Pierer, L. Berardinelli, M. Felderer, S. Ristov, Twin2Clouds: Cost-aware digital twin engineering and deployment across federated clouds, in: 2025 ACM/IEEE 28th International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C), 2025, pp. 264–275. doi:10.1109/MODELS-C68889.2025.00045.
- [15] D. Souza, T. Webber, E. Wanner, Towards federated, autonomous and cognitive digital twins with darling, in: 2025 20th Annual System of Systems Engineering Conference (SoSE), 2025, pp. 1–6. doi:10.1109/SoSE66311.2025.11083783.
- [16] A. Ricci, A. Croatti, S. Montagna, Pervasive and connected digital twins—a vision for digital health, *IEEE Internet Computing* 26 (5) (2022) 26–32. doi:10.1109/MIC.2021.3052039.
- [17] M. Martinelli, J. Zhang, A.-K. Splettstoßer, M. Picone, M. Lippi, A. Wortmann, Hierarchical digital twin ecosystem for industrial manufacturing scenarios, in: 2024 50th Euromicro Conference on Software Engineering and Advanced Applications (SEAA), 2024, pp. 56–63. doi:10.1109/SEAA64295.2024.00018.
- [18] M. Gervasi, L. Bove, A. Corallo, M. Miglietta, L. Negro, G. Pellegrino, F. Ria, C. A. Visaggio, Systems of digital twins: A systematic literature review and a formal framework, *Digital Engineering* 9 (2026) 100100. doi:https://doi.org/10.1016/j.dte.2026.100100.
- [19] Y. Wu, K. Zhang, Y. Zhang, Digital twin networks: A survey, *IEEE Internet of Things Journal* 8 (18) (2021) 13789–13804. doi:10.1109/JIOT.2021.3079510.
- [20] P. G. Larsen, P. Talasila, J. Fitzgerald, Towards the Composition of Digital Twins, Springer Nature Switzerland, Cham, 2024, pp. 103–122. doi:10.1007/978-3-031-67114-2_5.
- [21] P. Calvo-Bascones, A. Voisin, P. Do, M. A. Sanz-Bobi, A collaborative network of digital twins for anomaly detection applications of complex systems. snitch digital twin concept, *Computers in Industry* 144 (2023) 103767. doi:https://doi.org/10.1016/j.compind.2022.103767.
- [22] C. Vergara-Marcillo, R. Bahsoon, N. Tziritas, G. Theodoropoulos, A connectionist approach to federated digital twins, in: M. H. Lees, W. Cai, S. A. Cheong, Y. Su, D. Abramson, J. J. Dongarra, P. M. A. Sloot (Eds.), Computational Science – ICCS 2025, Springer Nature Switzerland, Cham, 2025, pp. 60–74.
- [23] A. Ricci, A. Croatti, S. Mariani, S. Montagna, M. Picone, Web of digital twins, *ACM Trans. Internet Technol.* 22 (4) (Nov. 2022). doi:10.1145/3507909.
- [24] A. Giulianelli, S. Burattini, A. Ciorte, A. Ricci, Engineering interoperable ecosystems of digital twins: A web-based approach, in: Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems, MODELS Companion '24, Association for Computing Machinery, New York, NY, USA, 2024, p. 476–485. doi:10.1145/3652620.3688263.
- [25] Z. Yang, Z. Wu, M. Luo, W.-L. Chiang, R. Bhardwaj, W. Kwon, S. Zhuang, F. S. Luan, G. Mittal, S. Shenker, I. Stoica, SkyPilot: An intercloud broker for sky computing, in: Symposium on Networked Systems Design and Implementation (NSDI 23), USENIX, Boston, MA, 2023, pp. 437–455.
- [26] S. Ristov, S. Pedratscher, T. Fahringer, AFCL: An Abstract Function Choreography Language for serverless workflow specification, *Fut. Gen. Comp. Syst.* 114 (2021) 368 – 382.
- [27] B. Combemale, J. Kienzle, G. Mussbacher, H. Ali, D. Amyot, M. Bagherzadeh, E. Batot, N. Bencomo, B. Benni, J.-M. Bruel, J. Cabot, B. H. Cheng, P. Collet, G. Engels, R. Heinrich, J.-M. Jezequel, A. Koziol, S. Mosser, R. Reussner, H. Sahraoui, R. Saini, J. Sallou, S. Stinckwich, E. Syriani, M. Wimmer, A hitchhiker's guide to model-driven engineering for data-centric systems, *IEEE Software* 38 (4) (2021) 71–84. doi:10.1109/MS.2020.2995125.